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Lab4

Part A – Basic grep and Filtering

Display all lines containing the word failed.
No word as failed starting with lowercase f.

Display all lines containing failed, ignoring case.

```
msis@master:~/Lab4$ grep -i "failed" security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:29:58] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:17] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
[2025-08-18 08:31:03] Failed login for user admin from 172.31.200.9 port 2222
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:38] Failed login for user admin from 198.51.100.77 port 2222
```

Display all lines that contain either Failed or Denied.

```
msis@master:~/Lab4$ grep -Ei "Failed|Denied" security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:15:44] ERROR Connection denied from 198.51.100.77 port 9001
[2025-08-18 08:29:58] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:17] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
[2025-08-18 08:31:03] Failed login for user admin from 172.31.200.9 port 2222
[2025-08-18 08:32:14] ERROR Connection from 1.1.1.1 denied on port 3389
[2025-08-18 08:37:12] WARNING upload attempt denied for user guest
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:38] Failed login for user admin from 198.51.100.77 port 2222
```

Display all lines that do not contain the word INFO.
grep -v "INFO" security-log.txt

Display all lines containing the exact word root.
grep -w "root" security_log.txt

Display all occurrences of the word attack along with their line numbers.
grep -n "attack" security_log.txt

Count the number of lines containing ERROR.

```
msis@master:~/Lab4$ grep -c "ERROR" security_log.txt
10
```

Display only the filenames of files whose contents contain the word malware, assuming multiple log files are available.

```
msis@master:~/Lab4$ grep -I "malware" *.txt
[2025-08-18 08:18:07] WARNING Request to http://malware-test.example.org/download.php from
 10.20.30.40
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.o
rg/evil.exe
[2025-08-18 08:45:26] Suspicious file malware.exe executed by user guest
[2025-08-18 08:45:49] ERROR file malware.exe hash MD5=5d41402abc4b2a76b9719d911017c592
Suspicious malware file evil.exe
```

Display the first five lines containing Suspicious.

```
msis@master:~/Lab4$ grep "Suspicious" security_log.txt|head -5
[2025-08-18 08:16:05] Suspicious connection from 198.51.100.77 to evilc2.ddns.net:9001
[2025-08-18 08:16:19] Suspicious outbound request https://evilc2.ddns.net/update.php
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.o
rg/evil.exe
[2025-08-18 08:19:45] Suspicious file invoice_2025.pdf downloaded from https://phishing-ex
ample.com/invoice_2025.pdf
[2025-08-18 08:20:51] Suspicious email phishing.alert@evilmail.ru contained malicious atta
chment
```

Display the last five failed-login entries.

```
msis@master:~/Lab4$ grep -i "failed" security_log.txt|tail -5
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
[2025-08-18 08:31:03] Failed login for user admin from 172.31.200.9 port 2222
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:38] Failed login for user admin from 198.51.100.77 port 2222
```

Part B – Regular Expression Extraction

Regular Expression Extraction is the process of using a specific text pattern (a regular expression or "regex") to search through a body of text and pull out matching pieces of information

1. **Extract all IPv4 addresses from the log.**

```
msis@master:~/Lab4$ grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' security_log.txt
192.168.10.5
192.168.10.21
203.0.113.45
203.0.113.45
203.0.113.45
203.0.113.45
10.10.5.23
198.51.100.77
198.51.100.77
192.168.10.21
10.20.30.40
203.0.113.99
10.1.2.3
10.1.2.3
10.1.2.3
10.1.2.3
172.16.5.10
172.31.200.9
192.168.1.50
192.168.10.5
8.8.8.8
192.168.10.5
1.1.1.1
203.0.113.45
192.168.10.21
192.168.10.21
203.0.113.99
203.0.113.99
198.51.100.77
198.51.100.77
198.51.100.77
198.51.100.77
198.51.100.77
198.51.100.77
8.8.8.8
203.0.113.45
```

2. Extract all IP addresses belonging to the 10.x.x.x private network.

```
msis@master:~/Lab4$ grep -oE '\b10(\.[0-9]{1,3}){3}\b' security_log.txt
10.10.5.23
10.20.30.40
10.1.2.3
10.1.2.3
10.1.2.3
10.1.2.3
```

3. Extract all addresses belonging to the 192.168.x.x network.

```
msis@master:~/Lab4$ grep -oE '\b192\.168(\.[0-9]{1,3}){2}\b' security_log.txt
192.168.10.5
192.168.10.21
192.168.10.21
192.168.1.50
192.168.10.5
192.168.10.5
192.168.10.21
192.168.10.21
```

4. **Extract all email addresses.**

```
msis@master:~/Lab4$ grep -oE '[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}' security_log.
txt
attacker123@phishing-example.com
security.team@example.com
phishing.alert@evilmail.ru
student2025@university.in
```

5. **Extract only email addresses whose domain ends in .com.**

```
msis@master:~/Lab4$ grep -oE '[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.com\b' security_log.txt
attacker123@phishing-example.com
security.team@example.com
```

6. **Extract email addresses whose username contains at least one digit.**

```
msis@master:~/Lab4$ grep -oE '[A-Za-z0-9._%+-]*[0-9][A-Za-z0-9._%+-]*@[A-Za-z0-9.-]+\.[A-Za-z]{2,}' security_log.txt
attacker123@phishing-example.com
student2025@university.in
```

7. **Extract all URLs beginning with http:// or https://.**

```
msis@master:~/Lab4$ grep -Eo 'https?://[^[:space:]]+' security_log.txt
https://evilc2.ddns.net/update.php
http://malware-test.example.org/download.php
http://malware-test.example.org/evil.exe
https://phishing-example.com/invoice_2025.pdf
https://target.example.com/login.php
https://www.example.com/index.html
https://shop.example.com/admin.php
http://legacy.example.net/login.php
https://portal.university.in/login.php
https://updates.bad-domain.ru/update.php
https://updates.bad-domain.ru/update.php
http://evil.example.net/dropper.exe
https://files.example.com/upload.php
https://command.bad-domain.in/backdoor.exe
```

8. **Extract only HTTPS URLs.**

```
msis@master:~/Lab4$ grep -Eo 'https://[^[:space:]]+' security_log.txt
https://evilc2.ddns.net/update.php
https://phishing-example.com/invoice_2025.pdf
https://target.example.com/login.php
https://www.example.com/index.html
https://shop.example.com/admin.php
https://portal.university.in/login.php
https://updates.bad-domain.ru/update.php
https://updates.bad-domain.ru/update.php
https://files.example.com/upload.php
https://command.bad-domain.in/backdoor.exe
```

9. Extract URLs ending with .php.

```
msis@master:~/Lab4$ grep -Eo 'https?://[^[:space:]]+\.php' security_log.txt
https://evilc2.ddns.net/update.php
http://malware-test.example.org/download.php
https://target.example.com/login.php
https://shop.example.com/admin.php
http://legacy.example.net/login.php
https://portal.university.in/login.php
https://updates.bad-domain.ru/update.php
https://updates.bad-domain.ru/update.php
https://files.example.com/upload.php
```

10. Extract all CVE identifiers in the form:CVE-YYYY-NNNN

```
msis@master:~/Lab4$ grep -Eo 'CVE-[0-9]{4}-[0-9]{4}' security_log.txt
CVE-2025-1234
CVE-2024-3094
CVE-2025-9876
CVE-2023-4448
CVE-2025-5432
CVE-2025-5432
```

11. Extract only CVEs reported for the year 2025.

```
msis@master:~/Lab4$ grep -Eo 'CVE-2025-[0-9]{4}' security_log.txt
CVE-2025-1234
CVE-2025-9876
CVE-2025-5432
CVE-2025-5432
```

12. Extract all 32-character hexadecimal MD5 hashes.

```
msis@master:~/Lab4$ grep -Eo '\b[[:xdigit:]]{32}\b' security_log.txt
44d88612fea8a8f36de82e1278abb02f
e2fc714c4727ee9395f324cd2e7f331f
5d41402abc4b2a76b9719d911017c592
```

13. Extract all 40-character hexadecimal SHA-1 hashes.

```
msis@master:~/Lab4$ grep -Eo '\b[[:xdigit:]]{40}\b' security_log.txt
3395856ce81f2b7382dee72602f798b642f14140
```

14. Extract all 64-character hexadecimal SHA-256 hashes.

```
msis@master:~/Lab4$ grep -Eo '\b[[:xdigit:]]{64}\b' security_log.txt
275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f
3a7bd3e2360a3d80a9d0b7b93d7f90d7c0b8c9a3e9f1d2c4b5a6978877665544
```

15. Extract all timestamps having the format:
YYYY-MM-DD HH:MM:SS
grep -Eo '[0-9]{4}-[0-9]{2}-[0-9]{2} [0-9]{2}:[0-9]{2}:[0-9]{2}' security_log.txt

Part C – Anchors and Regex Patterns
1. Display all lines beginning with a timestamp.
grep -E '^\[[0-9]{4}-[0-9]{2}-[0-9]{2}' security_log.txt

2. Display all lines beginning with the word Suspicious.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep '^Suspicious' security_log.txt
Suspicious activity detected from 203.0.113.45
Suspicious malware file evil.exe
```

3. Display lines ending with a numeric value.
grep '[0-9]$' security_log.txt

4. Display lines beginning with ERROR or WARNING.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -E '^(ERROR|WARNING)' security_log.txt
ERROR service terminated with code 137
WARNING connection retry 3
```

5. Display lines whose first character is a digit.
grep '^[0-9]' security_log.txt

6. Find lines ending in .exe.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep '\.exe$' security_log.txt
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org/evil.exe
[2025-08-18 08:24:25] Suspicious SHA256 hash=275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f
 file payload.exe
[2025-08-18 08:25:18] WARNING file payload.exe stored at C:\Users\Public\Downloads\payload.exe
[2025-08-18 08:35:58] Suspicious download http://evil.example.net/dropper.exe to C:\Temp\dropper.exe
[2025-08-18 08:47:04] Suspicious download of file backdoor.exe from https://command.bad-domain.in/backdoor.ex
e
Suspicious malware file evil.exe
```

7. Find lines containing filenames ending in .exe, .sh, or .pdf.
grep -E '\.(exe|sh|pdf)\b' security_log.txt

8. Identify lines containing the exact word root, but not words such as rooted.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -w 'root' security_log.txt
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:27:24] Suspicious crontab created for user root
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
```

9. Display blank lines in the file.
grep '^$' security_log.txt

10. Display all nonblank lines.
grep -v '^$' security_log.txt

Part D – Linux Filters and Pipelines

1. Extract all IPv4 addresses, sort them, and remove duplicates.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt | sort -u
10.10.5.23
10.1.2.3
10.20.30.40
1.1.1.1
172.16.5.10
172.31.200.9
192.168.10.21
192.168.10.5
192.168.1.50
198.51.100.77
203.0.113.45
203.0.113.99
8.8.8.8
```

2. Count the number of unique IPv4 addresses present in the log.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt | sort -u | wc -l
13
```

3. Extract all IP addresses and display them in sorted order.
grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b|([0-9a-fA-F]{1,4}:){2,7}[0-9a-fA-F]{1,4}' security_log.txt | grep -vE '^[0-9]{2}:[0-9]{2}:[0-9]{2}$' | sort

4. Determine which IP addresses appear more than once.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt | sort | uniq -d
10.1.2.3
192.168.10.21
192.168.10.5
198.51.100.77
203.0.113.45
203.0.113.99
8.8.8.8
```

5. Determine the IP address that occurs most frequently.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt | sort | uniq -c | sort -nr | head -n 1
      7 198.51.100.77
```

6. Extract all usernames from authentication entries and display unique usernames.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -iE 'user [a-zA-Z0-9_]+' security_log.txt | awk '{print $2}' | sort -u
08:11:14]
08:12:03]
08:12:10]
08:12:17]
08:14:33]
08:21:15]
08:27:24]
08:29:58]
08:30:17]
08:30:44]
08:31:03]
08:37:12]
08:42:01]
08:42:19]
08:42:38]
08:45:26]
08:46:37]
```

7. Count the number of authentication attempts associated with each username.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -iE 'user [a-zA-Z0-9_]+' security_log.txt | awk '{print $2}' | sort | uniq -c
      1 08:11:14]
      1 08:12:03]
      1 08:12:10]
      1 08:12:17]
      1 08:14:33]
      1 08:21:15]
      1 08:27:24]
      1 08:29:58]
      1 08:30:17]
      1 08:30:44]
      1 08:31:03]
      1 08:37:12]
      1 08:42:01]
      1 08:42:19]
      1 08:42:38]
      1 08:45:26]
      1 08:46:37]
```

8. Extract all HTTP status codes and determine how many times each status code occurs.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE 'status [0-9]{3}' security_log.txt | awk '{print $2}' | sort | uniq -c
      4 200
      1 302
      1 403
      2 404
      2 500
```

9. Display the five most frequently occurring IP addresses.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt | sort | uniq -c | sort -nr | head -n 5
      7 198.51.100.77
      6 203.0.113.45
      4 192.168.10.21
      4 10.1.2.3
      3 203.0.113.99
```

10. Display the three most frequently occurring usernames.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -iE 'user [a-zA-Z0-9_]+' security_log.txt | awk '{print $2}' | sort | uniq -c | sort -nr | head -n 3
      1 08:46:37]
      1 08:45:26]
      1 08:42:38]
```

11. Extract all destination ports and display them in numerical order.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE 'port [0-9]+' security_log.txt | awk '{print $2}' | sort -n
22
22
22
22
443
2222
2222
2222
2222
2222
3389
4444
4444
4444
8080
8080
8443
9001
```

12. Remove duplicate port numbers.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE 'port [0-9]+' security_log.txt | awk '{print $2}' | sort -nu
22
443
2222
3389
4444
8080
8443
9001
```

13. Count the number of unique destination ports.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE 'port [0-9]+' security_log.txt | awk '{print $2}' | sort -u | wc -l
8
```

14. Extract all file extensions appearing in the log and determine their frequency.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE '\.(exe|sh|pdf|php|html)\b' security_log.txt | sort | uniq -c
     14 .exe
      1 .html
      8 .pdf
      9 .php
      5 .sh
```

15. Find the five most frequently occurring domains.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE '\b[a-zA-Z0-9.-]+\.(com|org|net|in|ru)\b' security_log.txt | sort | uniq -c | sort -nr | head -n 5
      3 evilc2.ddns.net
      2 updates.bad-domain.ru
      2 portal.university.in
      2 phishing-example.com
      2 malware-test.example.org
```

Part E – Combining grep Conditions

1. Display lines containing both an IP address and the word Failed.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep 'Failed' security_log.txt | grep -E '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b'
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:29:58] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:17] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
[2025-08-18 08:31:03] Failed login for user admin from 172.31.200.9 port 2222
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:38] Failed login for user admin from 198.51.100.77 port 2222
```

2. Display lines containing both root and Failed.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep 'Failed' security_log.txt | grep 'root'
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
```

3. Display lines containing either download or upload.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -iE 'download|upload' security_log.txt
[2025-08-18 08:18:07] WARNING Request to http://malware-test.example.org/download.php from 10.20.30.40
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org/evil.exe
[2025-08-18 08:19:04] INFO file report.pdf saved to /home/analyst/downloads/report.pdf
[2025-08-18 08:19:45] Suspicious file invoice_2025.pdf downloaded from https://phishing-example.com/invoice_2025.pdf
[2025-08-18 08:25:18] WARNING file payload.exe stored at C:\Users\Public\Downloads\payload.exe
[2025-08-18 08:35:30] Suspicious download https://updates.bad-domain.ru/update.php to /tmp/update.sh
[2025-08-18 08:35:58] Suspicious download http://evil.example.net/dropper.exe to C:\Temp\dropper.exe
[2025-08-18 08:36:47] INFO Upload completed to https://files.example.com/upload.php
[2025-08-18 08:37:12] WARNING upload attempt denied for user guest
[2025-08-18 08:47:04] Suspicious download of file backdoor.exe from https://command.bad-domain.in/backdoor.exe
```

4. Display suspicious entries containing either .exe or .sh.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -i 'suspicious' security_log.txt | grep -E '\.(exe|sh)\b'
[2025-08-18 08:18:20] Suspicious file download evil.exe from http://malware-test.example.org/evil.exe
[2025-08-18 08:24:01] Suspicious SHA1 hash=3395856ce81f2b7382dee72602f798b642f14140 file updater.sh
[2025-08-18 08:24:25] Suspicious SHA256 hash=275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f file payload.exe
[2025-08-18 08:26:05] Suspicious script stored at /tmp/updater.sh
[2025-08-18 08:35:30] Suspicious download https://updates.bad-domain.ru/update.php to /tmp/update.sh
[2025-08-18 08:35:58] Suspicious download http://evil.example.net/dropper.exe to C:\Temp\dropper.exe
[2025-08-18 08:45:26] Suspicious file malware.exe executed by user guest
[2025-08-18 08:47:04] Suspicious download of file backdoor.exe from https://command.bad-domain.in/backdoor.exe
Suspicious malware file evil.exe
```

5. Display lines containing a CVE identifier and an IP address.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep 'CVE-' security_log.txt | grep -E '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b'
[2025-08-18 08:22:03] ERROR Exploit attempt CVE-2025-12345 from 203.0.113.99
[2025-08-18 08:47:31] ERROR CVE-2025-54321 exploited from 8.8.8.8 port 8080
```

6. Display lines containing both a filename and a hash value.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -i 'file' security_log.txt | grep -i 'hash'
[2025-08-18 08:18:32] ERROR file evil.exe hash MD5=44d88612fea8a8f36de82e1278abb02f
[2025-08-18 08:24:01] Suspicious SHA1 hash=3395856ce81f2b7382dee72602f798b642f14140 file updater.sh
[2025-08-18 08:24:25] Suspicious SHA256 hash=275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f file payload.exe
[2025-08-18 08:24:46] INFO MD5 hash=e2fc714c4727ee9395f324cd2e7f331f file archive.pdf
[2025-08-18 08:36:19] ERROR file dropper.exe hash SHA256=3a7bd3e2360a3d80a9d0b7b93d7f90d7c0b8c9a3e9f1d2c4b5a6978877665544
[2025-08-18 08:45:49] ERROR file malware.exe hash MD5=5d41402abc4b2a76b9719d911017c592
```

7. Display entries containing an IP address followed by a port.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -E '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b(:| +port +)[0-9]+' security_log.txt
[2025-08-18 08:10:01] INFO Server startup completed on 192.168.10.5 port 8080
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:15:44] ERROR Connection denied from 198.51.100.77 port 9001
[2025-08-18 08:29:11] INFO Connection from 10.1.2.3 port 22
[2025-08-18 08:29:32] INFO Connection from 10.1.2.3 port 22
[2025-08-18 08:29:58] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:17] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
[2025-08-18 08:31:03] Failed login for user admin from 172.31.200.9 port 2222
[2025-08-18 08:31:28] INFO Internal scan from 192.168.1.50 to 192.168.10.5 port 443
[2025-08-18 08:31:51] WARNING External scan from 8.8.8.8 to 192.168.10.5 port 8443
[2025-08-18 08:32:40] Suspicious connection 203.0.113.45:4444 detected
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:38] Failed login for user admin from 198.51.100.77 port 2222
[2025-08-18 08:47:31] ERROR CVE-2025-54321 exploited from 8.8.8.8 port 8080
```

8. Display HTTP requests containing both a URL and status code 404.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -E 'http|URL' security_log.txt | grep 'status 404'
[2025-08-18 08:33:55] WARNING URL https://shop.example.com/admin.php status 404
```

9. Display authentication failures but exclude entries generated from 127.0.0.1.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -iE 'fail|brute' security_log.txt | grep -v '127.0.0.1'
[2025-08-18 08:12:03] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:10] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:12:17] Failed login for user root from 203.0.113.45 port 4444
[2025-08-18 08:13:02] WARNING Multiple authentication failures from 203.0.113.45
[2025-08-18 08:29:58] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:17] Failed login for user guest from 10.1.2.3 port 22
[2025-08-18 08:30:44] Failed login for user admin from 172.16.5.10 port 2222
[2025-08-18 08:31:03] Failed login for user admin from 172.31.200.9 port 2222
[2025-08-18 08:42:01] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:19] Failed login for user root from 198.51.100.77 port 2222
[2025-08-18 08:42:38] Failed login for user admin from 198.51.100.77 port 2222
[2025-08-18 08:43:02] WARNING Brute force suspected from 198.51.100.77
```

10. Display attack-related entries but exclude informational (INFO) messages.

grep -vE 'INFO|DEBUG|SUCCESS' security_log.txt

Part F – Negative Matching

1. Extract all IPv4 addresses except those beginning with 127..

grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt | grep -v '^127\.'

2. Display all log entries except successful login events.

grep -vi "Successful login" security_log.txt

3. Display all entries that do not contain INFO, DEBUG, or SUCCESS.

grep -vE "INFO|DEBUG|SUCCESS" security_log.txt

4. Find HTTP requests whose status code is not 200.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -i "status" security_log.txt | grep -v "status
200"
[2025-08-18 08:33:55] WARNING URL https://shop.example.com/admin.php status 404
[2025-08-18 08:34:16] ERROR URL http://legacy.example.net/login.php status 500
[2025-08-18 08:35:04] WARNING URL https://updates.bad-domain.ru/update.php status 302
[2025-08-18 08:40:51] WARNING Request from 203.0.113.99 status 404
[2025-08-18 08:41:13] ERROR Request from 203.0.113.99 status 500
[2025-08-18 08:41:36] Suspicious request from 198.51.100.77 status 403
```

5. Display suspicious traffic excluding connections to 192.168.x.x.

grep -i "suspicious" security_log.txt | grep -v "192\.168\."

6. Extract email addresses except those belonging to the organization's domain.

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE '[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z
{2,}' security_log.txt | grep -v '@example\.com$'
attacker123@phishing-example.com
phishing.alert@evilmail.ru
student2025@university.in
```

Part G – Redirection

1. Extract all IPv4 addresses and save them into:
ips.txt

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt | s
ort | uniq -c | sort -nr
      7 198.51.100.77
      6 203.0.113.45
      4 192.168.10.21
      4 10.1.2.3
      3 203.0.113.99
      3 192.168.10.5
      2 8.8.8.8
      1 192.168.1.50
      1 172.31.200.9
      1 172.16.5.10
      1 10.20.30.40
      1 10.10.5.23
      1 1.1.1.1
```

2. Extract all email addresses and save them into:
emails.txt

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE 'CVE-[0-9]{4}-[0-9]+' security_log.txt | sort -u
CVE-2023-44487
CVE-2024-3094
CVE-2025-12345
CVE-2025-54321
CVE-2025-987654
```

3. Extract all URLs and save them into:
urls.txt

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE "(INFO|WARNING|ERROR|DEBUG|SUCCESS|Suspicious)" security_log.txt
| sort | uniq -c | sort -nr
     25 INFO
     24 Suspicious
     20 WARNING
     10 ERROR
      1 SUCCESS
      1 DEBUG
```

4. Extract all hashes and save them into:
hashes.txt
grep -i "Suspicious.*download" security_log.txt > suspicious_downloads.txt

5. Extract failed-login entries and save them into:
failed_login.txt

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep "User-Agent:" security_log.txt | sed -E 's/.*User-Agent: (.*)/\1/' | sort -u
curl/8.5.0
Mozilla/5.0 Chrome/139.0
Mozilla/5.0 EvilScanner/2.1
python-requests/2.31.0
```

6. Extract suspicious entries and append them to:
incident_report.txt

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -oE '\b[a-fA-F0-9]{32}\b|\b[a-fA-F0-9]{40}\b|\b[a-fA-F0-9]{64}\b' security_log.txt
 | sort -u
275a021bbfb6489e54d471899f7db9d1663fc695ec2fe2a2c4538aabf651fd0f
3395856ce81f2b7382dee72602f798b642f14140
3a7bd3e2360a3d80a9d0b7b93d7f90d7c0b8c9a3e9f1d2c4b5a6978877665544
44d88612fea8a8f36de82e1278abb02f
5d41402abc4b2a76b9719d911017c592
e2fc714c4727ee9395f324cd2e7f331f
```

7. Store unique IP addresses in:
Unique_ips.txt

```
sanjanarao@sanjanarao-VMware-Virtual-Platform:~/Assign4$ grep -i "Failed login" security_log.txt | sed -nE 's/.*user ([^ ]+) .*/\1/p' | sort | u
niq -c
      3 admin
      2 guest
      5 root
```

8. Store IP-frequency statistics in:
Ip_frequency.txt
grep -oE '\b([0-9]{1,3}\.){3}[0-9]{1,3}\b' security_log.txt | sort | uniq -c | sort -nr > Ip_frequency.txt

9. Redirect normal output to results.txt and errors to errors.txt.
grep 'ERROR' security_log.txt > results.txt 2> errors.txt

10. Extract all domains from the log entries containing Domain accessed, convert the domain names to lowercase, sort them alphabetically, remove duplicates, and save the result in domains.txt.
grep -oP 'Domain accessed \K\S+' security_log.txt | tr 'A-Z' 'a-z' | sort -u > domains.txt

11. Identify all IP addresses involved in failed-login attempts. Convert the output into a clean list containing one unique IP per line and save it as failed_login_ips.txt.
grep -i 'failed login' security_log.txt | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort -u > failed_login_ips.txt

Combining filters

12. Extract all failed-login usernames and save them into:
failed_users.txt
grep -i 'failed login' security_log.txt | grep -oP 'user \K\S+' | sort -u > failed_users.txt

13. Extract all domains and save them in lowercase into:
domains.txt
grep -oP 'Domain accessed \K\S+' security_log.txt | tr 'A-Z' 'a-z' | sort -u > domains.txt

14. Extract all MD5 hashes, convert them to uppercase, and save them into:
Md5_hashes.txt
grep -oE '\b[a-fA-F0-9]{32}\b' security_log.txt | tr 'a-z' 'A-Z' > md5_hashes.txt

15. Extract source IP addresses from failed-login records, sort them, remove duplicates, and redirect the result to:

failed_ips.txt
grep -i 'failed login' security_log.txt | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort -u > failed_ips.txt

16. Create a file named clean_log.txt in which all square brackets are removed from the original log.
tr -d '[]' < security_log.txt > clean_log.txt

17. Create normalized_log.txt by converting the complete log file to lowercase.
tr 'A-Z' 'a-z' < security_log.txt > normalized_log.txt

18. Extract all domain names, convert them to lowercase, sort them, remove duplicates, and save them to unique_domains.txt.
grep -oP 'Domain accessed \K\S+' security_log.txt | tr 'A-Z' 'a-z' | sort -u > unique_domains.txt

19. Find all failed-login entries and extract only the usernames.
grep -i 'failed login' security_log.txt | grep -oP 'user \K\S+'

20. Find all failed-login entries, extract only the source IP addresses, and display them in sorted order.
grep -i 'failed login' security_log.txt | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort

21. Extract IP addresses associated with failed logins and determine the unique IPs.
grep -i 'failed login' security_log.txt | grep -oE '([0-9]{1,3}\.){3}[0-9]{1,3}' | sort -u

22. Extract usernames involved in failed logins, convert them to uppercase, and remove duplicate values.
grep -i 'failed login' security_log.txt | grep -oP 'user \K\S+' | tr 'a-z' 'A-Z' | sort -u

23. Find all lines containing Domain accessed, extract only the domain names, and convert them to lowercase.
grep 'Domain accessed' security_log.txt | grep -oP 'Domain accessed \K\S+' | tr 'A-Z' 'a-z'

24. Find all lines containing MD5=, extract only the MD5 hash values, and convert hexadecimal letters to uppercase.
grep 'MD5=' security_log.txt | grep -oP 'MD5=\K[a-fA-F0-9]{32}' | tr 'a-z' 'A-Z'

25. Find entries containing User-Agent:, extract the user-agent field, and convert it to lowercase.
grep 'User-Agent:' security_log.txt | grep -oP 'User-Agent: \K.*' | tr 'A-Z' 'a-z'

26. Find all CVE-related lines, extract the CVE identifiers, and convert them to lowercase.
grep -oE 'CVE-[0-9]{4}-[0-9]{4,}' security_log.txt | tr 'A-Z' 'a-z'

27. Find all entries containing .exe, extract the relevant filename, and convert it to uppercase.
grep -oE '[A-Za-z0-9_-]+\.exe' security_log.txt | sort -u | tr 'a-z' 'A-Z'

28. Extract all domains from Domain accessed entries, convert them to uppercase, sort them, and remove duplicates.

```
grep -oP 'Domain accessed \K\S+' security_log.txt | tr 'a-z' 'A-Z' | sort -u
```